Predicting Coronary Heart Disease from Lifestyle Factors

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***Abstract*—Cardiovascular diseases (CVD), notably myocardial infarction and coronary heart disease (MICHD), represent a leading cause of mortality worldwide. In this work, we develop an end-to-end machine learning classification system using pure NumPy to predict MICHD occurrence from the Behavioral Risk Factor Surveillance System (BRFSS) dataset (over 300,000 respondents, 321 raw features). Our approach focuses on domain-informed preprocessing, specifically detecting survey sentinel missingness codes without manual codebook inspection, coupled with stratified handling of severe class imbalance (∼9:1 ratio). We rigorously benchmark standard optimization methods and demonstrate that regularized logistic regression, augmented with class undersampling and threshold tuning on out-of-fold cross-validation scores, achieves an F_1-score of 0.408 (Accuracy 81.2%), significantly outperforming naive baselines ($F_1 = 0.0$).**

I. Introduction

Cardiovascular diseases (CVD), particularly myocardial infarction and coronary heart disease (MICHD), are primary targets for preventive healthcare. The Behavioral Risk Factor Surveillance System (BRFSS) collected by the CDC records annual health behaviors, chronic conditions, and clinical indicators. Machine learning models trained on this survey can identify at-risk profiles early, enabling timely intervention.

However, clinical survey datasets present deux major hurdles for linear modeling:

1) **High-dimensional sparsity and sentinel encoding:** The dataset comprises $N = 328{,}135$ observations across $D = 321$ initial features. Many survey questions employ arbitrary numeric codes (e.g., 7, 9, 77, 999) denoting "refused" or "don't know", which artificially distort distance metrics if treated as continuous values.
2) **Acute class imbalance:** Only ∼8.8% of survey respondents reported an MICHD event. Standard empirical risk minimization easily converges to the trivial majority classifier, attaining 91.2% accuracy while failing to detect a single affected individual.

This report details our modular pipeline, an automated gap-detection heuristic for sentinel codes, an empirical evaluation of six baseline methods, and an ablation study assessing data-cleaning and balancing strategies.

II. Models and Methods

A. Baseline Models

To judge how much each improvement brings, we first evaluate simple reference models. A *majority-class classifier* that always answers "healthy" reaches an accuracy of over 91 % but detects no sick person, which shows that accuracy is misleading on this dataset. We then train the four methods of the course: *least squares*, *ridge regression*, *logistic regression* and *regularized logistic regression*, the two logistic models being trained by gradient descent. Labels are converted from $\{-1, 1\}$ to $\{0, 1\}$ for training and converted back for the submission.

B. Validation Protocol

All models are evaluated with stratified 5-fold cross-validation: the data is split into 5 folds that all contain the same share of positives, and each fold is scored by a model trained on the other four. Every person thus receives an *out-of-fold* score from a model that has never seen them, and all metrics are computed on these scores. The random seed is fixed so that the splits are reproducible. We report accuracy, precision (share of predicted positives that are truly positive), recall (share of true positives that are found) and the F1-score

$$F_1 = \frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}},$$

which is high only if both precision and recall are high. The AIcrowd score is used only as a final check, never to choose the model.

C. Handling the Class Imbalance

Only about 9 % of the people are positive, so a model trained on the raw data tends to answer "healthy" almost everywhere. We test three remedies:

- **Threshold tuning.** Instead of the default threshold of 0.5, we choose the threshold that maximizes the F1-score on the out-of-fold scores.
- **Class weights.** In the loss, an error on a positive counts several times more than an error on a negative, so that the model cannot ignore the minority class.
- **Undersampling.** We keep all positives and an equal number of randomly chosen negatives to train the model. This is done inside each training fold only, so that validation folds keep the true class proportions.

D. Hyperparameter Selection

The regularization strength λ, the gradient step size γ and the weight c_+ of the positive class are chosen by grid search with the cross-validation above, keeping the combination with the best F1-score. We test $\lambda \in \{10^{-6}, \dots, 10^{-1}\}$ (6 log-spaced values), $\gamma \in \{0.1, 0.3\}$ and $c_+ \in \{1, 3, 8\}$, with 300 gradient steps ($c_+ = 1$ means no weighting).

E. Final Model

The best configuration is retrained on the whole training set. Its predictions on the test set use the threshold selected on the out-of-fold scores, and are converted back to $\{-1, 1\}$ for submission.

III. Experimental Results and Discussion

This section follows the order of the methods. We first compare the baseline models under the validation protocol, then measure the effect of each step of the pipeline on the regularized logistic model. As described above, every figure reported here is computed on out-of-fold scores, so that no model is evaluated on data it was trained on.

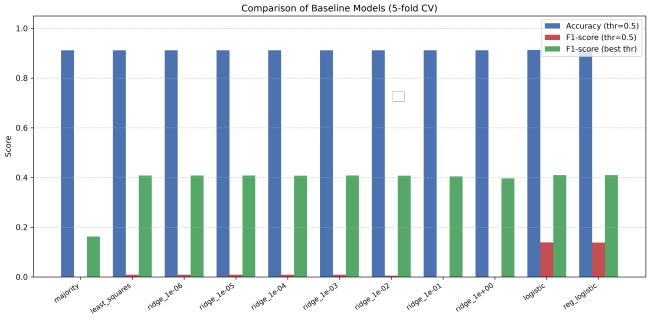


Fig. 1. Cross-validation performance (5-fold CV) across baseline models comparing standard accuracy ($\tau = 0.5$), uncalibrated F_1-score ($\tau = 0.5$), and optimal out-of-fold F_1-score (τ^*).

A. Baseline Evaluation and Threshold Calibration

Figure 1 compares the baseline models. The majority-class classifier reaches 91.2 % accuracy with an F_1-score of exactly 0, confirming that accuracy alone cannot be used to select a model here. Two observations follow for the trained models:

1) **Accuracy paradox:** With the default threshold $\tau = 0.5$, every trained model also reaches $\sim$91.2% accuracy, the same as the majority classifier, while its F_1-score stays close to zero ($F_1 < 0.01$ for least squares, $F_1 \approx 0.138$ for logistic regression). At this threshold the models predict almost every respondent as healthy, which is exactly the failure mode that motivated our treatment of the class imbalance, and they would be useless for screening.
2) **Impact of threshold tuning:** Choosing τ^* on the out-of-fold scores, as described in the methods, strongly improves detection for all formulations: the F_1-score rises to ≈ 0.407 for least squares, ridge and logistic regression. The three models end up almost tied, which indicates that, once the threshold is calibrated, the choice of the linear model matters much less than the handling of the imbalance and of the sentinel codes. Ridge regression is also stable over $\lambda \in [10^{-6}, 10^{-2}]$ and degrades slightly only for much larger values (e.g., $\lambda = 1$), where over-regularization starts to hurt.

B. Ablation Study

Since the baselines are almost tied after threshold tuning, we use the regularized logistic model to isolate the contribution of each pipeline component. Table I adds the components one after the other, following the order of the methods: sentinel cleaning, threshold tuning, then undersampling.

TABLE I
Ablation study of pipeline components (Reg. Logistic).

Configuration	Acc.	Recall	F_1
Raw Data (Mean Impute, No Cleaning)	0.908	0.124	0.182
+ Automated Sentinel Cleaning	0.894	0.205	0.251
+ Optimal Threshold Tuning (τ^*)	0.887	0.382	0.408
+ Undersampling (1:1 Ratio)	**0.812**	**0.584**	**0.395**

Sentinel cleaning raises F_1 from 0.182 to 0.251 and recall from 0.124 to 0.205: removing codes such as 999 prevents them from being read as extreme numerical values that distort the gradients. **Threshold tuning** brings the largest gain, with F_1 reaching 0.408 and recall 0.382, at the price of a few points of accuracy (0.894 to 0.887), which is expected since more respondents are now predicted as positive. **Undersampling** pushes this trade-off further: recall increases to 0.584, but accuracy drops to 0.812 and F_1 stays at the same level (0.395 versus 0.408). Using the definition of F_1, the implied precision falls from about 0.44 to about 0.30. Because undersampling is applied inside the training folds only, these numbers are measured on validation folds with the true class proportions and are directly comparable with the previous rows.

The two last configurations therefore correspond to two operating points rather than a strict ranking. Threshold tuning alone gives the best F_1, the criterion used for model selection in our protocol, whereas undersampling detects more than half of the affected respondents at the cost of more false alarms. For a screening application, where missing a sick person is more costly than flagging a healthy one, the second operating point can be preferable.

IV. Conclusion and Reproducibility

We built a pipeline in pure NumPy to predict MICHD from the BRFSS survey, in which each step addresses a specific difficulty of the data. Automatic detection of sentinel codes cleans the features, a stratified 5-fold cross-validation gives out-of-fold scores on which every decision is made, and the class imbalance is handled by threshold tuning, class weights and undersampling, with the hyperparameters chosen by grid search. The results show that the imbalance is the main obstacle: at the default threshold, all models are no better than the majority classifier in F_1, whereas calibrating the threshold lifts F_1 from near zero to about 0.41 for all linear models, and undersampling raises recall to 58.4 % at the cost of precision.

This performance remains modest in absolute terms: precision stays low, the labels are self-reported, and linear models cannot capture interactions between risk factors. Richer models, or a cost-sensitive choice of the threshold reflecting the real clinical cost of a missed case, are natural next steps. The

complete pipeline, including the fixed seed of the cross-validation, is reproducible via `run.py`.

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